EXPERIMENT - 75

Question 1

1. Problem Statement

Assign explicit values starting from 10 in an enumeration and print them.
Input: Nil
Output: A=10, B=11, C=12

2. Algorithm

Step 1: START
Step 2: Declare enum Letter with A = 10. B and C follow it, so their values are 11 and 12.
Step 3: Display A, B and C with their integer values.
Step 4: STOP

3. Test Case Table

#	Input	Expected Output	Actual Output	Result
1	Nil	A=10, B=11, C=12	A=10, B=11, C=12	**PASS**

4. Source Code

enum75a.c

```
/* Experiment 75, Question 1: explicit values in an enumeration. */
#include <stdio.h>

enum Letter { A = 10, B, C };

int main(void)
{
    printf("A=%d, B=%d, C=%d\n", A, B, C);
    return 0;
}
```

Question 2

1. Problem Statement

Use enumeration to represent menu choices (ADD, SUBTRACT, MULTIPLY) and perform operations using switch.
Input: ADD 10 20
Output: 30

2. Algorithm

Step 1: START
Step 2: Declare enum Operation with ADD, SUBTRACT and MULTIPLY.
Step 3: Read the operation name and the two numbers a and b. Change the letters of the name to capital letters.
Step 4: If the name is ADD, SUBTRACT or MULTIPLY, set op to that value. Otherwise display Invalid operation and stop.
Step 5: Use switch on op: for ADD set result = a + b, for SUBTRACT set result = a − b, and for MULTIPLY set result = a × b.
Step 6: Display result.
Step 7: STOP

3. Test Case Table

#	Input	Expected Output	Actual Output	Result
1	ADD 10 20	30	30	**PASS**
2	SUBTRACT 20 5	15	15	**PASS**
3	MULTIPLY 6 7	42	42	**PASS**
4	add 4 5	9	9	**PASS**
5	SUBTRACT 5 20	-15	-15	**PASS**
6	DIVIDE 10 2	Invalid operation	Invalid operation	**PASS**

4. Source Code

calc75.c

```
/* Experiment 75, Question 2: menu choices as an enumeration and a switch. */
#include <stdio.h>
#include <string.h>
#include <ctype.h>

enum Operation { ADD, SUBTRACT, MULTIPLY };

int main(void)
{
    char name[20];
    int a, b, i, result;
    enum Operation op;

    printf("Enter the operation and two numbers (ADD, SUBTRACT or MULTIPLY): ");
    if (scanf("%19s %d %d", name, &a, &b) != 3) {
        printf("Invalid input\n");
        return 1;
    }
    for (i = 0; name[i] != '\0'; i++)
        name[i] = toupper((unsigned char)name[i]);

    if (strcmp(name, "ADD") == 0)
        op = ADD;
    else if (strcmp(name, "SUBTRACT") == 0)
        op = SUBTRACT;
    else if (strcmp(name, "MULTIPLY") == 0)
        op = MULTIPLY;
    else {
        printf("Invalid operation\n");
        return 1;
    }

    switch (op) {
    case ADD:
        result = a + b;
        break;
    case SUBTRACT:
        result = a - b;
        break;
    case MULTIPLY:
```

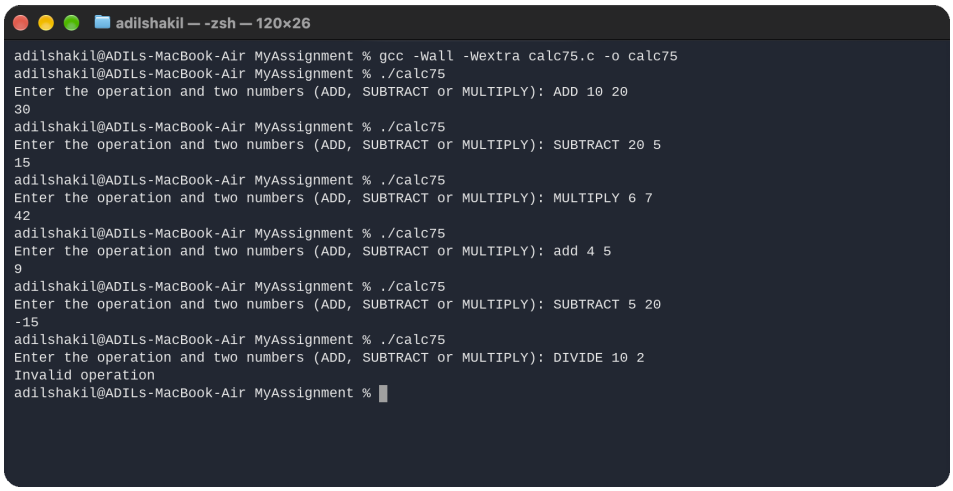
```
        result = a * b;
        break;
    }
    printf("%d\n", result);
    return 0;
}
```

5. Compilation and Output

Question 1: compiled with `gcc -Wall -Wextra enum75a.c -o enum75a`



Question 2: compiled with `gcc -Wall -Wextra calc75.c -o calc75`



Both programs built clean, and the first test of each is the example from the question.